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# Hybrid Ensemble Model Approaches for Stock Price Forecasting Using LSTM, Random Forest, ARIMA, and Linear Regression as Meta-Learner

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**Abstract**

Predicting stock market price remains a complex and challenging task due to the influence of various financial, economic, and behavioral factors. To address these challenges, numerous machine learning and deep learning techniques such as ARIMA, Random Forest, and Long Short-Term Memory (LSTM) have been proposed. However, each model exhibits inherent strengths and limitations in capturing specific data characteristics. This study proposes a hybrid ensemble framework that integrates ARIMA for linear pattern, Random Forest, for non-linear dependencies, and LSTM for temporal dynamics. A meta-learning strategy based on Linear Regression is implemented to perform weighted ensembling of individual model outputs. The experiment results across five major stocks demonstrate that the proposed approach significantly outperforms standalone models. The hybrid ensemble model achieved low MAPE scores of 0.5590 for AAPL, 0.7126 for XOM, and 5.944 for TSLA. Additionally, the MAE results across three stocks were 0.8155 to 33.82. The RMSE results were from 1.027 to 40.86. And lastly, the hybrid ensemble model achieved high R2 scores across the three stocks with range from 0.9304 to 0.9601. These findings highlight the effectiveness and robustness of the hybrid ensemble approach in stock price forecasting.

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*Keywords:* Stock Price Forecasting; Hybrid Ensemble Models; LSTM; Random Forest; ARIMA; Meta-Learning; Financial Time Series

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## 1. Introduction

Predicting stock market prices remains the most challenging task due to the complexity and volatility of financial systems, which are influenced by several interacting factors. These various factors include the broader business environment, business performance, and other physical or psychological factors [1]. The challenges of accurate prediction were further amplified during the global disruption such as COVID-19 pandemic, where traditional models often failed to generalize due to increased volatility and market irrationality. More recent studies indicate that model performance varied significantly under crisis conditions, necessitating more adaptive and hybrid approaches [16]. Consequently, the high complexity and instability of market behavior make stock price prediction difficult for individuals to accurately predict when to buy or sell stocks to maximize profit.

Machine learning (ML) and deep learning (DL) have been extensively used by researchers to make a stock price prediction model. Among classical machine learning approaches, the Auto Regressive Integrated Moving Average or in short (ARIMA) is mainly used in stock price prediction model. The ARIMA model has three main parts: p, d, q where p represents the order of AR, q represents the order of MA term and d represents order of differencing [4]. Although ARIMA is considered as the best model for baseline predicting stock prices model, past study shows that ARIMA model struggles to capture nonlinear data, and the complex dependencies present in stock price, making it less effective for volatile stock prices [2].

The Random Forest model (RF) is introduced in the stock price prediction field because RF can capture non-linear trends better than ARIMA. Random Forest model uses the combination of decision trees and then combines the result of each decision tree to reduce variance and increase accuracy. Because of that, RF is better in handling high dimension data and capturing non-linear data [3]. Recent study shows that RF can achieve around 80 to 90 percent in terms of stock price prediction [4]. Study also shows that combining RF with extra trees can increase the accuracy in stock price prediction [9]. Despite high accuracy for prediction, the Random Forest model has its weakness which is unable to capture temporal dependency in stock. Therefore, RF struggles with stock prices that have high temporal dependencies.

Besides ARIMA and RF, the Long Short-Term Memory model (LSTM) is gaining popularity in stock price prediction field. Past study shows that LSTM can predict stock price with more than 90 percent accuracy and capture some unpredictable movement due to global events such as Covid-19 [4]. The LSTM model is effective in capturing temporal dependency in stock price because it is a specialized type of recurrent neural network [21]. Another study also shows that combining LSTM and RF in an ensemble model can increase prediction accuracy across diverse stock prices [10]. Despite the strengths of the LSTM model, LSTM has some weaknesses such as requiring intensive computation and huge amounts of data to reach the best result.

A recent study shows another method for optimizing the result of stock price prediction. The study shows that by using a meta-learning the model can optimize itself even if its input with different data or shifts to a new task. The recent study shows that a meta-learning model called Meta-Stock is able to adjust the model's prediction strategy by learning the combination of the outputs from level 1 learners [12].

In this study, we propose a hybrid ensemble approach that integrates ARIMA, RF, LSTM, and meta-learner with linear regression. ARIMA is used to capture linear trends in stock prices. RF is used to capture non-linear relationships. LSTM is used to capture temporal dependencies. And lastly, the meta-learner is used to weigh each model output based on past performance. With this implementation, we hope that the proposed model can generalize stock prices more accurately.

## 2. Methodology

### *2.1. Research Design*

Our research employs a quantitative approach in exploring the performance of the hybrid machine learning for predicting the movement of stock prices. We selected three models for our hybrid machine learning, Long Short-Term Memory (LSTM), Random Forest (RF), and Autoregressive Integrated Moving Average (ARIMA). The study is based on the work of other researchers, which highlights the advantage of integrating statistical and machine learning models in financial forecasting [1].

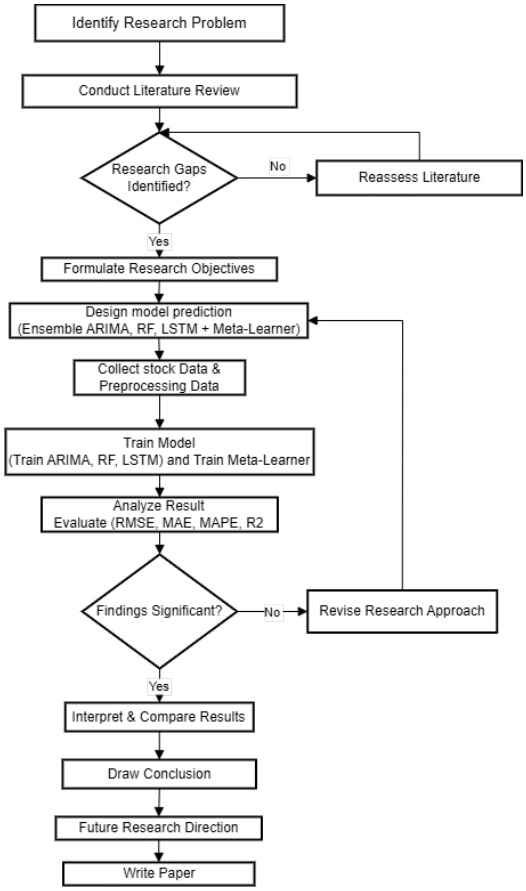



Fig. 1. Illustrate overall research flow of the proposed hybrid ensemble model.

### *2.2. Data Collection*

The dataset used consists of historical daily stock price data from January 1, 2010 to December 31, 2024. This data was retrieved from Yahoo Finance through a web scraping technique using the Python yfinance library. The complete dataset used in this study is publicly available and can be accessed online [22]. The features in the dataset include opening price, closing price, highest price, lowest price, and trading volume. We selected 5 different companies, namely Apple (AAPL), Tesla (TSLA), Exxon Mobil (XOM), Goldman Sachs (GS), and JPMorgan Chase (JPM), which come from different sectors such as technology, energy, and finance. These stocks are chosen so that we can test the generalizability of the hybrid model to various stock behaviors. The results from this study will help evaluate the robustness of the forecasting model to varying conditions. The time span of the data provides a sufficient amount of data for both training and validation. A good data structure is claimed to significantly improve the accuracy of machine learning-based financial forecasting [5].

### *2.3. Data Preprocessing*

To optimize the performance of the model, the data is preprocessed through several stages. First, missing values are handled using interpolation or forward fill techniques to ensure data continuity. Next, standardization is applied using a standard scaler to maintain data consistency. The dataset is then divided into 80% training data and 20% testing data to effectively evaluate the performance of the model. In addition, time series decomposition is applied to analyze

trends, seasonality, and random fluctuations in the data. Proper data preprocessing plays an important role in stock market prediction, as improper treatment of missing data or noise can greatly affect the accuracy of the model [6].

### *2.4. Model Implementation*

#### *2.4.1. Long Short-Term Memory (LSTM)*

In this research, we implemented the Long Short-Term Memory (LSTM) model to handle the sequential and time-dependent nature of stock price data. LSTM is a type of recurrent neural network (RNN) that is specifically designed to learn patterns across time. We used a two-layer stacked LSTM architecture, each with 64 units, followed by a dense layer to produce the final output. The model takes a sequence of the past 30 days (look-back window) as input to predict the next closing price. All input features were normalized using MinMaxScaler to improve training stability. The model was trained for 20 epochs with a batch size of 32, using the Adam optimizer and mean squared error (MSE) as the loss function. This setup allows the model to effectively learn from temporal data while maintaining generalization. The model consist of these following hyperparameters:

Table 1. LSTM Hyperparameters.

| Hyperparameter | Value | Description |
|---|---|---|
| units (1st layer) | 64 | Number of neurons in the first LSTM layer. |
| return_sequences | True | Returns the output sequence to the next LSTM layer. |
| input_shape | (look_back, num_features) | Shape of the input sequences, defined dynamically. |
| units (2nd layer) | 64 | Number of neurons in the second LSTM layer. |
| optimizer | Adam | Optimizer used to minimize the loss during training. |
| loss | mean_squared_error | Loss function used to measure prediction error. |
| epochs | 20 | Number of complete passes through the training dataset. |
| batch_size | 32 | Number of samples processed before the model is updated. |

LSTM can capture long-range dependencies in the trends in the stock price very well, which is an important component in hybrid forecasting models [4].

#### *2.4.2. Random Forest*

Random Forest (RF) was selected as one of the base models due to its ability to capture complex non-linear relationships in stock price movements. It works by constructing an ensemble of decision trees and averaging their outputs, which helps reduce overfitting and improve robustness. In our configuration, the model used 100 decision trees (n_estimators = 100) and a fixed random seed (random_state = 42) for reproducibility. The input features for RF

included several technical indicators such as moving averages (SMA), MACD, volatility, and volume averages. This model serves as a reliable option for modeling high-dimensional, non-linear patterns that are common in financial data. The model includes the following hyperparameters:

Table 2. Random Forest Hyperparameters.

| Hyperparameter | Value | Description |
|---|---|---|
| n_estimators | 100 | Number of trees in the forest. |
| random_state | 42 | Seed value for reproducibility. |

Studies have established that Random Forest handles big data in the stock market with great efficiency with minimal overfitting relative to linear models [7]. Moreover, a recent study shows that random forest's comparative advantage over LSTM in shorter time horizons and lower complexity tasks [20].

*2.4.3. ARIMA*

To capture the linear components of stock price trends, we implemented the ARIMA model, a traditional yet widely used statistical method for time series forecasting. The model was set with an order of (4, d, 0), where p=4 represents the number of autoregressive terms, d is the differencing order determined through the Augmented Dickey-Fuller (ADF) test for stationarity, and q=0 indicates no moving average component. This dynamic setting for d allows the model to adapt to the stationarity level of each stock's time series. ARIMA is especially useful for modeling short-term trends and serves as a solid baseline in our ensemble framework. The model consists of these following hyperparameters:

Table 3. ARIMA Hyperparameters.

| Hyperparameter | Value | Description |
|---|---|---|
| order | (4, d, 0) | ARIMA order where: p = 4 (autoregressive), d = chosen based on ADF test, q = 0 |
| d | Determined by ADF test | The differencing order for stationarity. |

ARIMA remains a strong benchmark for short-term stock price forecasting [1].

*2.5. Hybrid Model Development*

We develop our hybrid models by integrating ARIMA with LSTM and Random Forest with LSTM, aiming to maximize the strengths of both statistical and machine learning methodologies. The hybrid configurations are as follows:

- LSTM-ARIMA: ARIMA handles the long-range trends, while LSTM handles the short-range changes
- LSTM-Random Forest: The temporal features are deeply extracted by LSTM, with Random Forest providing robustness by eliminating overfitting.

Empirical data show hybrid approaches outperform standalone approaches by reconciling statistical power with the power of deep learning [1]. In addition, a study also confirms that combining multiple models leads to better generalization across volatile stocks [17]. Moreover, a recent study demonstrated that hybrid ensemble models

integrating statistical and machine learning methods achieved significantly higher predictive accuracy in financial time series forecasting compared to individual models [11].

### *2.6. Model Evaluation*

The models are evaluated based on the following performance metrics:

*2.6.1. Mean Absolute Error (MAE): Measures the average absolute prediction error.*

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (1)$$

Where '$y_i$' is the original closing price. '$\hat{y}_i$' refers to the predicted closing price. And lastly, 'n' represents the total length of the window.

*2.6.2. Root Mean Squared Error (RMSE): Quantifies the square root of the average squared differences.*

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (2)$$

Where '$y_i$' is the original closing price. '$\hat{y}_i$' refers to the predicted closing price. And lastly, 'n' represents the total length of the window.

*2.6.3. Mean Squared Percentage Error (MAPE): Evaluates prediction accuracy as a percentage.*

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \quad (3)$$

Where '$y_i$' is the original closing price. '$\hat{y}_i$' refers to the predicted closing price. And lastly, 'n' represents the total length of the window.

*2.6.4. R-squared (R2): Determines the proportion of variance explained by the model.*

$$R^2\ Score = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2} \quad (4)$$

Where '$y_i$' is the original closing price. '$\hat{y}_i$' refers to the predicted closing price and '$\bar{y}_i$' represents the mean of the original closing price. Lastly, 'n' represents the total length of the window.

The evaluation follows best practices in stock market forecasting [6].

### *2.7. Limitations*

Potential restrictions include external factors that affect market volatility and stock prices. Sudden dramatic changes in stock prices caused by events such as economic or political conditions can affect the model's accuracy and reliability. Additionally, the computational complexity involved in training hybrid models poses a challenge, as it requires significant computing power and time, making the process slower and more resource intensive. A recent studies have also found that the heteroscedasticity and irregular volatile patterns of RNN and LSTM-based approaches are still a challenge [18]. Despite these limitations, this approach ensures a comprehensive evaluation of hybrid machine learning models for stock price forecasting.

## 3. Result

### 3.1. Quantitative Performance

To test the effectiveness of the hybrid ensemble model, a comparison is made between ARIMA, Random Forest, LSTM, and the Hybrid ensemble itself across five companies such as Apple (AAPL), Exxon Mobil (XOM), Goldman Sachs (GS), JP Morgan (JPM), and Tesla (TSLA). The predicted closing price will be examined using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Square Percentage Error (MAPE), and R2 Score.

Table 4. Prediction Scores for Apple (AAPL)

| Stock Model | AAPL ARIMA | AAPL Random Forest | AAPL LSTM | AAPL Hybrid (Weighted) |
|---|---|---|---|---|
| MAE | 12.79 | 9.911 | 1.885 | **1.356** |
| MAPE | 5.203 | 4.033 | 0.7843 | **0.5590** |
| RMSE | 15.33 | 11.94 | 2.418 | **1.759** |
| $R^2$ | -2.028 | -0.8360 | 0.9247 | **0.9601** |

Table 5. Prediction Scores for Goldman Sachs (GS)

| Stock Model | GS ARIMA | GS Random Forest | GS LSTM | GS Hybrid (Weighted) |
|---|---|---|---|---|
| MAE | 11.97 | 22.32 | 9.466 | **5.940** |
| MAPE | 2.059 | 3.778 | 1.646 | **1.027** |
| RMSE | 14.75 | 26.00 | 12.50 | **8.083** |
| $R^2$ | -0.04550 | -2.248 | 0.2486 | **0.6862** |

Table 6. Prediction Scores for JP Morgan (JPM)

| Stock Model | JPM ARIMA | JPM Random Forest | JPM LSTM | JPM Hybrid (Weighted) |
|---|---|---|---|---|
| MAE | 4.239 | 6.282 | 2.778 | **1.746** |
| MAPE | 1.761 | 2.596 | 1.153 | **0.7306** |
| RMSE | 5.160 | 7.178 | 3.627 | **2.489** |
| $R^2$ | -0.1136 | -1.154 | 0.4498 | **0.7409** |

Table 7. Prediction Scores for Exxon Mobil (XOM)

| Stock Model | XOM ARIMA | XOM Random Forest | XOM LSTM | XOM Hybrid (Weighted) |
|---|---|---|---|---|
| MAE | 5.935 | 2.807 | 1.487 | **0.8155** |
| MAPE | 5.414 | 2.457 | 1.286 | **0.7126** |
| RMSE | 7.589 | 3.192 | 1.792 | **1.027** |
| $R^2$ | -1.315 | 0.5902 | 0.8708 | **0.9575** |

Table 8. Prediction Scores for Tesla (TSLA)

| Stock Model | TSLA ARIMA | TSLA Random Forest | TSLA LSTM | TSLA Hybrid (Weighted) |
|---|---|---|---|---|
| MAE | 280.1 | 66.32 | 139.4 | **33.82** |
| MAPE | 54.20 | 12.14 | 24.23 | **5.944** |
| RMSE | 319.9 | 80.53 | 154.9 | **40.86** |

| | | | | |
|---|---|---|---|---|
| $R^2$ | -3.263 | 0.7298 | -0.00005101 | **0.9304** |

All the data above were taken from training the model for short term prediction.

Based on the tables above, the model that achieved high R2 scores is considered to be the best prediction model. The proposed hybrid ensemble model outperformed the individual models ARIMA, Random Forest, and LSTM across all five stocks. The Hybrid Ensemble achieved high R2 scores on XOM, TSLA, and AAPL. However, the proposed model showed moderate performance on JPM and GS. In addition, LSTM model had high scores on AAPL and XOM. In contrast, Random Forest worked well on TSLA and slightly performed well on XOM but performed poorly on AAPL, GS, and JPM. Additionally, ARIMA consistently produced negative values of R2 scores across all stocks, indicating its ineffectiveness for forecasting highly volatile and non-linear stocks like TSLA and AAPL. Overall, the proposed model significantly surpasses all base models across all evaluation metrics.

### *3.2. Visualization of Predictions*

To visually evaluate the model's performance, we plotted the actual vs. predicted closing prices for each stock across the five companies. The following figures demonstrate the forecasting performance of ARIMA, Random Forest, LSTM, and the Hybrid Ensemble model on selected stocks.




Fig. 2. (a) True Values vs Hybrid Prediction Closing Price for APPL; (b) True Values vs Hybrid Prediction Closing Price for GS; (c) True Values vs Hybrid Prediction Closing Price for JPM; (d) True Values vs Hybrid Prediction Closing Price for XOM; (e) True Values vs Hybrid Prediction Closing Price for TSLA.

## 4. Conclusion

In this study, we have implemented ARIMA, Random Forest, and LSTM as the first level learner and linear regression as the second level learner as a stock price prediction model. The model was evaluated in terms of MAE, MSE, RMSE, and R2 across five stocks such as APPL, GS, JPM, TSLA, and XOM. Based on the result, the proposed model performed better compared to the other singular model.

The proposed model uses meta-learning as the second level learner. The meta-learner acts as the learner that weighs the output of each model based on previous prediction. Even though one of the models can perform worse than the other model, the meta-learner can weigh each output so that the predicted price is closer to the actual price.

For future reference, there are several areas for future improvement. Firstly, the proposed model uses static weight assignment. Exploring dynamic weight learning by using neural meta-learners or attention-based mechanisms can increase the prediction accuracy. Secondly, expanding model diversity by implementing transformer-based architectures, or graph neural networks could enhance the ensemble's predictive capabilities. Furthermore, XGBoost Extra Trees, and convolutional architectures like Conv-RNN have also been proposed as effective alternatives for capturing non-linearities and price momentum patterns [14][19][20].

## Data Availability

The Authors demonstrate our commitment to data openness and transparency. To facilitate further research, we have made the data used in this paper publicly available. The data used by the authors can be opened via the link below:
https://drive.google.com/drive/folders/1_CydDOQ1BS7LC0tfzDeWqys2tFaNJj15?usp=sharing

## Author Contribution Statement

The authors would like to express their sincere gratitude to all contributors who played a crucial role in the successful completion of this research paper. Special thanks are extended to our academic supervisors, Irene Anindaputri Iswanto and Setiawan Joddy, whose guidance and constructive feedback throughout the project have been invaluable. This paper is the result of a collaborative effort among all authors, with each member contributing significantly to the study:

- Brainardo Kenneth Siahu contributed to the design and implementation of the methodology, dataset acquisition, and development of the predictive models.
- Fernando Darren Rachman was responsible for conducting the literature review, co-developing the methodology section, supporting the model implementation, and preparing supplementary materials including the video documentation.
- Wisnutomo Adji Nugroho led the writing of the abstract, introduction, result analysis, and conclusion. He also contributed to dataset collection and model coding efforts.
- Irene Anindaputri Iswanto and Setiawan Joddy served as academic advisors, providing comprehensive guidance, reviewing drafts, and ensuring the academic integrity and scientific relevance of the work.

All authors have read and approved the final version of the manuscript.

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